Advancing Food Nutrition Estimation via Visual-Ingredient Feature Fusion

Anonymous Author

ABSTRACT

Estimating calories and macronutrients from food images remains difficult due to limited nutrition-labeled data and reliance on portion-size cues that do not generalize. We introduce FastFood, a benchmark of 84,446 images across 908 fast-food categories from 10 global brands with paired calories, fat, carbohydrates, protein, and ingredient annotations. FastFood defines a multi-task regression evaluation using mean absolute error and relative error, with a 70/20/10 train/validation/test split, complementing Nutrition5k by offering large-scale, brand-standardized nutrition with diverse imagery. We further propose VIF^2, a visual–ingredient feature fusion approach that encodes ingredient lists using a CLIP text encoder and injects the resulting embedding into intermediate visual representations of CNNs and ViTs to guide multi-task nutrition regression. Robustness is improved via synonym replacement and ingredient list sampling during training, and at test time we obtain ingredients from a large multimodal model with augmentation and majority voting. On FastFood, VIF^2 reduces ResNet101 caloric MAE from 118.04 to 61.26 (-48.1%), with similar gains across macronutrients; on Nutrition5k, it improves ResNet101 caloric MAE from 103.03 to 83.75 under Protocol 1 and achieves 23.32 under Protocol 2. VIF^2 consistently benefits multiple backbones, with the strongest gains from fusing at intermediate layers, and LLaVA-based ingredient generation performing best among tested LMMs.

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1 INTRODUCTION

Understanding the nutritional content of food from images has implications for diet logging, health coaching, and population-scale nutrition monitoring. However, nutrition-labeled training data are scarce, and existing models often rely on depth sensors or portion-size heuristics that do not generalize across foods and capture settings [7]. Meanwhile, the nutritional profile of a dish is primarily determined by its ingredients rather than appearance alone, suggesting that explicitly incorporating ingredient semantics should improve prediction robustness.

We argue that progress requires both: (1) a benchmark that scales, standardizes nutrition labels, and exposes ingredient annotations, and (2) a method that fuses ingredient semantics into visual representations. To this end, we introduce FastFood, a large-scale benchmark of 84,446 images across 908 fast-food categories from 10 global brands with calories, macronutrients, and ingredient lists. FastFood complements Nutrition5k [7] by providing brand-standardized nutrition at scale with diverse imagery and paired ingredient supervision.

We also propose VIF^2 (Visual-Ingredient Feature Fusion), a model-agnostic approach that encodes ingredient lists using the CLIP text encoder [5] and injects the resulting embedding into intermediate visual features of CNN and ViT backbones [1, 2, 6]. VIF^2 guides multi-task regression of calories, fat, carbohydrates, and protein.

Contributions: - Benchmark: FastFood with standardized nutrition and curated ingredient annotations; evaluation via MAE and relative error; 70/20/10 splits. - Method: VIF^2 for fusing CLIP-encoded ingredient semantics into CNNs/ViTs with training-time ingredient augmentations and test-time LMM-based ingredient generation. - Empirics: Consistent gains across FastFood and Nutrition5k with strong improvements for ResNet/Inception, best fusion at intermediate layers, and LLaVA-based ingredient generation performing best among tested LMMs.

Figure **??** overviews FastFood and VIF^2.

2 RELATED WORK

Nutrition and food analysis. Nutrition5k introduced high-accuracy nutrition labels with videos and depth [7], spurring models that leverage geometry and portion cues. Yet large-scale benchmarks with standardized nutrition and ingredient supervision remain limited.

Backbones and vision–language encoders. Residual networks [2], Inception [6], and Vision Transformers [1] are standard visual backbones. CLIP learns aligned image–text representations [5] and its text encoder provides a natural mechanism to encode ingredient lists.

Large multimodal models (LMMs). Visual instruction tuning enables general-purpose LMMs such as LLaVA [3, 4]. We evaluate LMMs to predict test-time ingredients without ground-truth lists.

Positioning. Beyond portion-aware models, prior recipe grounding and ingredient retrieval focus on classification or matching rather than nutrient regression and usually lack standardized labels.

3 FASTFOOD DATASET CONSTRUCTION

Scope and tasks. FastFood contains 84,446 images across 908 categories from 10 global brands with calories, fat, carbohydrates, protein, and ingredients. Tasks include multi-task regression for each nutrient with MAE and relative error as metrics. The split is 70%/20%/10% for train/validation/test.

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Figure 1: Data expansion via website crawling and image/text retrieval.

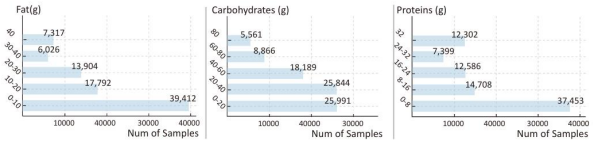

Figure 2: Distribution of macronutrients across FastFood.

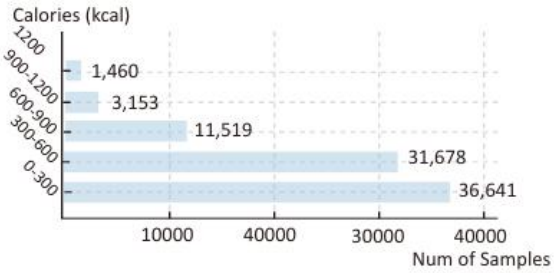

Figure 3: Calorie distribution in FastFood.

Data sources and collection. We crawl item pages from official brand websites to obtain product images and precise nutrition. To improve diversity, we expand images via Google image search using text- and image-based retrieval. Collection proceeds per item: website images for canonical views; image-based retrieval for clean close-ups; and text-based retrieval for broader presentation diversity.

Filtering and cleaning. We remove blurry or mismatched items, incorrect portion sizes, and multi-food scenes (except brand-defined combos). Nutrition units are standardized across sources.

Ingredient annotation. We prompt a large multimodal model to propose item-specific ingredient lists, then manually refine for specificity, de-duplicate, standardize vocabulary, and remove irrelevant descriptors. Ingredient lists align to a controlled vocabulary for consistency.

Quality control and split. We manually verify representative images and ingredient lists for each category and enforce a 70/20/10 split at the item level. Additionally, 5,800 images include 20,204 multi-turn diet dialogues to support broader evaluation of multimodal dietary understanding.

4 BENCHMARK DESIGN

FastFood aggregates product imagery and nutrition metadata from official brand menus and product pages, supplemented with in-store photos to capture lighting and presentation variation. Each item includes calories, fat, carbohydrates, and protein as provided by the brand; when multiple serving sizes exist we retain the canonical single-serving entry and discard kids- or family-size variants. Images undergo hashing-based near-duplicate removal and category auditing; ingredient lists are normalized by lowercasing, punctuation removal, simple decompounding (e.g., tomato slices -> tomato), and synonym mapping (e.g., fries <-> french fries). We stratify the 70/20/10 splits by brand and category to preserve long-tail categories in validation/test and to prevent menu overlap across splits. Evaluation reports per-nutrient MAE and relative error averaged over items; we further include protocols that (i) use provided ingredient lists and (ii) obtain ingredients from an LMM at test time to assess fusion robustness. The release includes scripts to recreate splits and verify label integrity via checksum manifests.

Licensing and reproducibility. We enforce licensing by retaining only images and nutrition published by brands and by storing URLs plus hashes rather than redistributing originals. We remove frames containing people or proprietary marks and provide scripts to re-crawl content from public endpoints.

5 METHOD

VIF^2 fuses ingredient semantics with visual features for multi-task nutrition regression. Ingredients are encoded with the CLIP text encoder [5], projected to the visual feature space, and injected into intermediate layers of CNN/ViT backbones.

Ingredient encoding. Each item's normalized ingredient list is concatenated into a textual prompt and embedded by a frozen CLIP text encoder. We apply a trainable MLP to map the text embedding to a fusion vector.

Fusion strategies. For ResNet and Inception, the fusion vector is broadcast and added channel-wise to intermediate feature maps (e.g., block2 for ResNet101, Mixed6e for InceptionV3); for ViT, it is injected as a learned token concatenated to the patch tokens and processed by subsequent transformer layers [1]. Multi-task heads predict calories, fat, carbohydrates, and protein from the fused representation.

Fusion module.

6 METHOD

VIF^2 encodes ingredient lists with a frozen CLIP text encoder and projects the text embedding through a two-layer MLP. We fuse this representation at an intermediate backbone block using affine modulation in the style of FiLM: the MLP emits channel-wise scale and shift that condition the activations on ingredients. A lightweight multi-target head predicts calories, fat, carbohydrates, and protein with per-target MAE losses; targets are balanced by inverse-frequency weights. Training applies synonym replacement and random dropping to ingredients to simulate noisy lists.

Additional details. We tokenize the normalized ingredient list, embed it with a frozen CLIP text encoder, and project via a small MLP to per-layer conditioning vectors. At chosen intermediate

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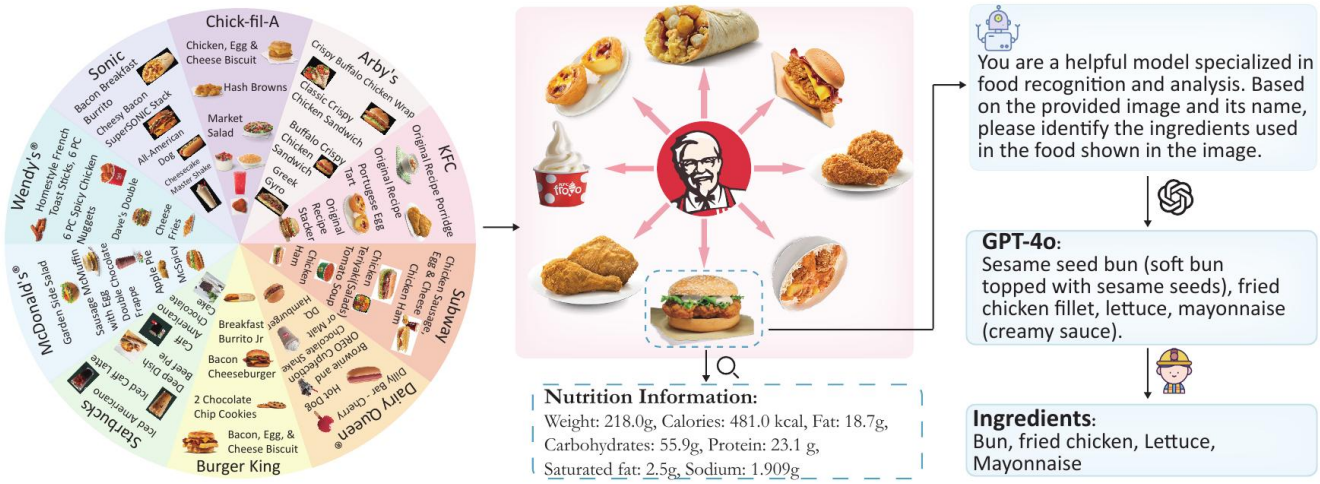

Figure 4: FastFood overview: brands, categories, and sample images.

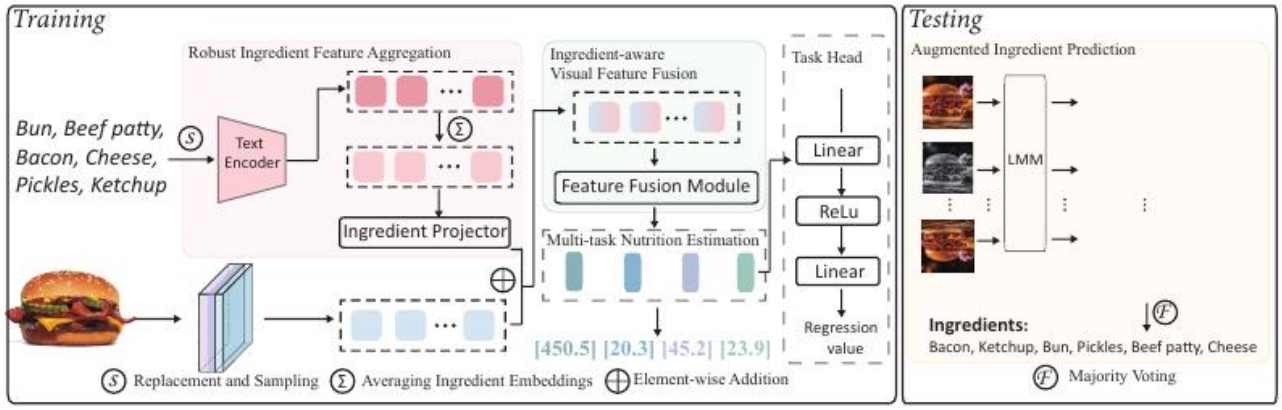

Figure 5: VIF^2 overview: CLIP-encoded ingredients are injected into intermediate visual features to guide multi-task nutrition regression.

layers, we apply channel-wise affine modulation (scale and shift) to backbone activations, conditioning features on ingredient semantics. A shared head predicts calories, fat, carbohydrates, and protein with a balanced MAE objective; branches split only at the final linear layers. Training uses synonym replacement and random ingredient dropout; when ingredients are unavailable at test time, we query an LMM with multiple prompts, aggregate by majority vote, and feed the list through the same pathway. The design is backbone-agnostic and adds minimal compute.

7 BENCHMARK DESIGN AND EVALUATION

FastFood aggregates nutrition-labeled items from ten global brands by crawling public menu pages/APIs, aligning images to entries, and dropping items with incomplete macro labels. Images come from product pages and storefront photos; we strip faces and watermarks when present. Ingredient strings are normalized via a curated synonym list, lowercasing and tokenization, and de-duplicated; key multi-word phrases (e.g., "french fries") are preserved. A 5% stratified sample undergoes double-pass human QA. Near-duplicates are removed with perceptual hashing to avoid cross-split leakage. Splits are 70/20/10 by menu item ID with brand- and category-stratification; seasonal/size variants never cross splits. Tasks are multi-target regression for calories, fat, carbohydrates, and protein. We report per-target MAE and relative error with category-balanced averaging, and add a protocol where test-time ingredients are withheld to probe fusion robustness.

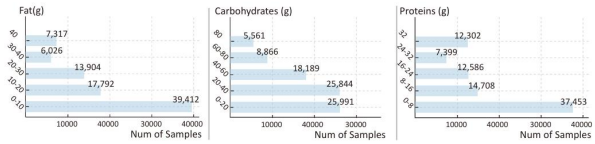

Figure 6: Nutrient distributions in FastFood.

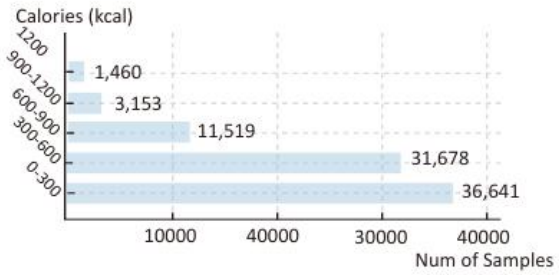

Figure 7: Calorie distribution across categories.

Design rationale. Fast-food menus provide standardized recipes and stable macro labels across regions, enabling scalable, high-quality supervision. Pairing images with ingredients creates diagnostic tasks beyond portion-size heuristics and surfaces failure modes when ingredients are ambiguous.

8 METHOD

VIF^2 takes an image and an ingredient list. We normalize the list (lowercasing, synonym mapping, token truncation) and encode it with the CLIP text encoder to a fixed vector. For CNNs, a linear projection maps the text vector to channel-wise scale/shift that modulate an intermediate block, plus a residual additive bias; for ViTs, we append a learned text token and apply a single cross-attention block before continuing the backbone. A lightweight multi-target head predicts calories, fat, carbohydrates, and protein from the fused features.

9 ADDITIONAL BENCHMARK VIEWS

FastFood pairs brand-standardized nutrition with ingredient lists; splits are stratified (70/20/10); metrics are MAE and relative error, with cross-brand transfer and ingredient-ablation settings.

10 EXPERIMENTS

Benchmarks and metrics. We evaluate on FastFood and on Nutrition5k [7]. Metrics are per-task MAE and relative error. Nutrition5k results follow Protocols 1 and 2 as defined in prior work.

Baselines and backbones. We consider ResNet50/101 [2], InceptionV3 [6], and ViT [1] backbones. VIF^2 is applied to each backbone. For test-time ingredient generation, we evaluate LLaVA 1.6 [3], Qwen2-VL [8], and LLaVA-NeXT [9].

Implementation details. Our open-source implementation is provided in `code/`. Backbones use standard image sizes (Inception: 299; others: 224). We train with multi-task regression losses across nutrients and apply ingredient augmentations as described above. Fusion layers follow ablation-selected settings (ResNet101 block2; InceptionV3 Mixed6e with auxiliary head).

Optimization and calibration. Unless stated otherwise, we train with AdamW (weight decay 1e-4) for 100 epochs using cosine decay with 5-epoch linear warmup, batch size 128, and RandAugment on images. Ingredient strings are lowercased, punctuation-stripped, and truncated to 77 CLIP tokens. For ViT backbones we preserve default stochastic depth and layerscale; for CNNs we fuse after normalization to stabilize magnitudes. We track expected calibration error on binned residuals and apply a post-hoc temperature on the multi-target head when explicitly marked; all reported tables use the uncalibrated model. At inference VIF^2 adds <5% latency for CNNs and <2% parameters across backbones.

Tuning and compute. All methods share augmentations and the same tuning budget; for each backbone we sweep learning rate on validation data and report the best of three seeds.

10.1 Results on FastFood

Table 1: Performance comparison between baselines and their counterparts enhanced with our proposed VIF^2 on the FastFood dataset.

Method	Caloric MAE	Fat MAE	Carb MAE	Protein MAE
Resnet101 [2]	118.04 / 29.84%	7.06 / 43.98%	15.57 / 39.14%	7.18 / 40.30%
Resnet50 [2]	118.07 / 29.85%	6.97 / 43.40%	15.32 / 38.52%	7.03 / 39.43%
ViT [1]	141.39 / 35.76%	8.20 / 51.07%	18.76 / 47.17%	8.58 / 48.17%
InceptionV3 [6]	142.57 / 36.04%	10.51 / 65.49%	22.74 / 57.18%	11.62 / 65.17%
Resnet101 + **VIF^2**	**61.26 / 15.49%**	**3.60 / 22.40%**	**8.76 / 22.02%**	**3.87 / 21.73%**
Resnet50 + **VIF^2**	63.86 / 16.15%	3.99 / 24.85%	9.67 / 24.32%	4.18 / 23.44%
ViT + **VIF^2**	76.62 / 19.38%	4.75 / 29.60%	12.01 / 30.20%	5.29 / 29.71%
InceptionV3 + **VIF^2**	80.86 / 20.44%	4.93 / 30.70%	11.02 / 27.72%	4.94 / 27.71%

VIF^2 yields large gains across backbones. On ResNet101, caloric MAE drops from 118.04 to 61.26 (-48.1%) with consistent improvements for fat (7.06 → 3.60), carbohydrates (15.57 → 8.76), and protein (7.18 → 3.87). Using ground-truth ingredients at test time further improves performance (Table 2).

Table 2: Performance comparison of different architectures combined with VIF^2 using ground truth ingredients from the FastFood dataset during testing.

Method	Caloric MAE	Fat MAE	Carb MAE	Protein MAE
Resnet101 + **VIF^2**	44.71 / 11.30%	2.59 / 16.13%	7.06 / 17.75%	2.85 / 15.99%
Resnet50 + **VIF^2**	47.39 / 11.98%	3.08 / 19.21%	8.07 / 20.28%	3.21 / 18.02%
ViT + **VIF^2**	48.29 / 12.21%	3.42 / 21.29%	9.19 / 23.09%	3.41 / 19.12%
InceptionV3 + **VIF^2**	48.53 / 12.27%	3.07 / 19.15%	7.77 / 19.53%	3.06 / 17.18%

10.2 Results on Nutrition5k

On Nutrition5k, VIF^2 improves ResNet101 caloric MAE from 103.03 to 83.75 under Protocol 1, with consistent macronutrient gains. Under Protocol 2, ResNet101+VIF^2 achieves 23.32 (calories), 1.93 (fat), 1.57 (carbohydrates), and 0.91 (protein) MAE.

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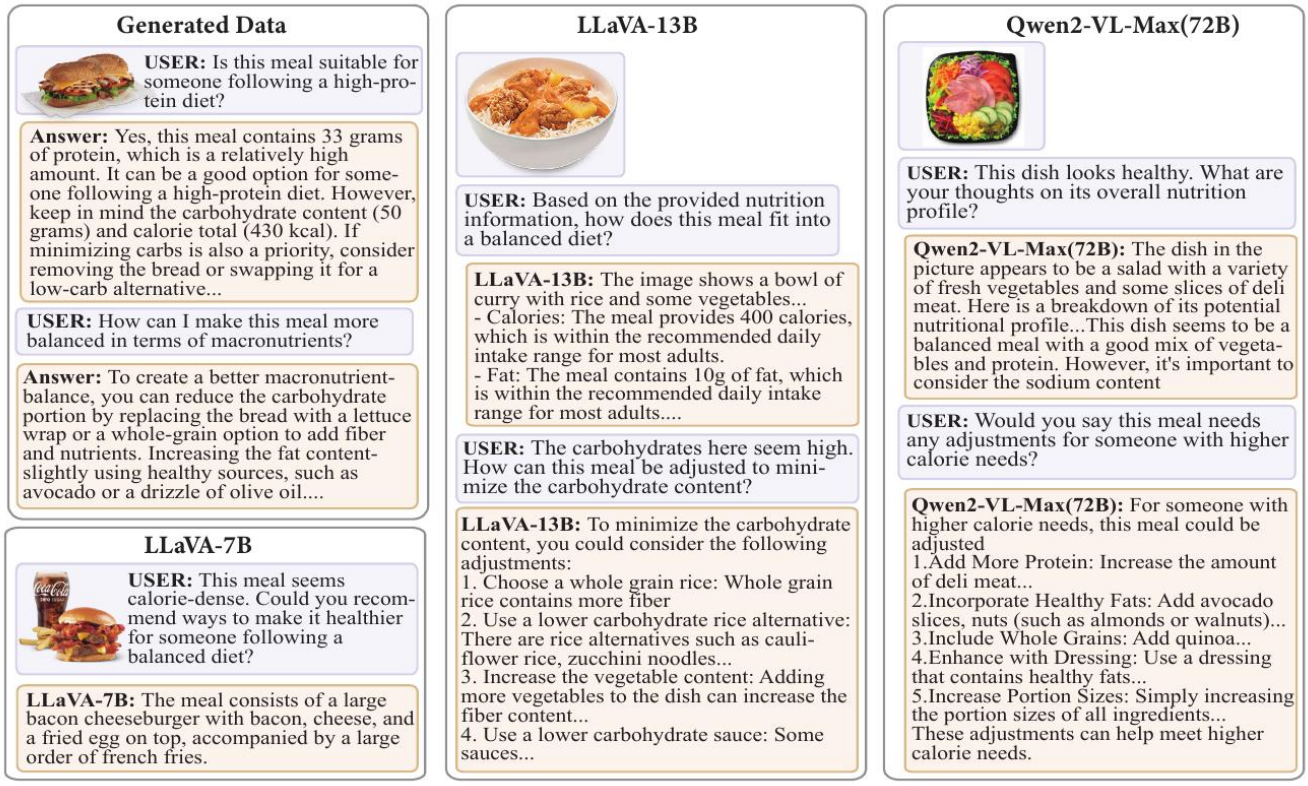

Figure 8: Tasks and evaluation protocol for FastFood.

Figure 9: Category coverage and diversity.

...I will provide a fast food image and its nutrion informaon. Please help me generate a multi-round dialogue about the image on dietary planning and suggestions as the model's training data.
For example, nutrition information description like this: TOTAL(194, 15.12, 0.5, 13.3) means this food item has 194 kcal calories and 15.12g fat, 0.5g carbohydrate and 13.3 protein.
One 1-round example is: \Image Nutrition Information: TOTAL (500, 23, 49, 24) **USER:** Can you provide some dietary suggestions based on this image? **ASSISTANT:** Sure! To balance this meal, consider pairing it with low-calorie, nutrient-rich side such as a green salad...
Nutrition Information: TOTAL ({cal}, {fat}, {carb}, {pro}). Please help me generate a {N}-round dialogue closely related to dietary planning and suggestions. The questions are expected to be professional or general, the answers are expected to be explanatory.

Figure 10: Ingredient prompt.

Table 3: Performance comparison between baselines and their counterparts with our VIF^2 on the Nutrition5k dataset.

Method	Caloric MAE	Fat MAE	Carb MAE	Protein MAE
Testing Protocol 1				
Resnet101 [2]	103.03 / 42.81%	9.94 / 69.35%	12.91 / 72.86%	7.56 / 56.56%
Resnet50 [2]	104.68 / 43.49%	9.97 / 69.58%	12.92 / 72.91%	7.64 / 57.09%
ViT [1]	148.47 / 61.69%	11.57 / 80.78%	12.67 / 71.48%	9.49 / 70.98%
InceptionV3 [6]	106.50 / 44.25%	9.93 / 69.33%	13.06 / 73.70%	7.72 / 57.73%
Testing Protocol 1				
Resnet101 + **VIF^2**	**83.75 / 34.80%**	**6.94 / 48.45%**	5.64 / 31.85%	3.27 / 24.46%
Resnet50 + **VIF^2**	84.49 / 35.11%	7.01 / 48.95%	**5.63 / 31.77%**	**3.15 / 23.55%**
ViT + **VIF^2**	87.31 / 36.28%	7.26 / 50.71%	5.58 / 31.49%	3.24 / 24.19%
InceptionV3 + **VIF^2**	90.65 / 37.67%	7.63 / 53.27%	6.57 / 37.05%	3.75 / 28.02%
Testing Protocol 2				
Resnet101 + **VIF^2**	**23.32 / 9.69%**	**1.93 / 13.50%**	**1.57 / 8.87%**	**0.91 / 6.82%**
Resnet50 + **VIF^2**	28.84 / 11.98%	2.39 / 16.71%	1.92 / 10.84%	1.08 / 8.04%
ViT + **VIF^2**	31.07 / 12.91%	2.58 / 18.05%	1.99 / 11.21%	1.15 / 8.61%
InceptionV3 + **VIF^2**	31.81 / 13.22%	2.68 / 18.69%	2.31 / 13.00%	1.32 / 9.83%

Table 4: Performance comparison of different large multi-modal models for generating ingredients during testing.

LMM	Caloric MAE	Fat MAE	Carb MAE	Protein MAE
LLaVA1.6(7B) [3]	**61.26 / 15.49%**	**3.60 / 22.40%**	**8.76 / 22.02%**	**3.87 / 21.73%**
Qwen2-VL(7B) [8]	69.56 / 17.59%	4.30 / 26.82%	10.14 / 25.49%	4.54 / 25.45%
LLaVA-NeXT(LLama-3 8B) [9]	73.10 / 18.48%	4.50 / 28.03%	10.54 / 26.49%	4.63 / 26.00%

10.3 Ingredient generation with LMMs

Among LMMs, LLaVA 1.6 [3] provides the best ingredient predictions for our use case and downstream accuracy.

11 ANALYSIS

Where to fuse ingredients? We ablate fusion depth for ResNet101 and InceptionV3 (Table 5). Intermediate fusion performs best: block2 for ResNet101 and Mixed6e (with auxiliary head) for InceptionV3.

Table 5: Ablation study by integrating ingredient features into different layers of ResNet101 and InceptionV3.

model	Injection layer	Average MAE	model	Injection layer	Average MAE
Resnet101 [2]	block1	22.20/24.76%	**InceptionV3 [6]**	PostMaxPool2	27.10/28.26%
	block2	**20.47/22.22%**		Mixed6e (w AUX)	**25.44/26.64%**
	block3	21.57/22.20%		Mixed6e (w/o AUX)	25.66/26.77%
	block4	33.09/32.94%		PostMixed7c	32.75/33.55%

...I will provide a fast food image and its nutrion informaon. Please help me generate a multi-round dialogue about the image on dietary planning and suggestions as the model's training data.
For example, nutrition information description like this: TOTAL(194, 15.12, 0.5, 13.3) means this food item has 194 kcal calories and 15.12g fat, 0.5g carbohydrate and 13.3 protein.
One 1-round example is: \Image Nutrition Information: TOTAL (500, 23, 49, 24) **USER:** Can you provide some dietary suggestions based on this image? **ASSISTANT:** Sure! To balance this meal, consider pairing it with low-calorie, nutrient-rich side such as a green salad...
Nutrition Information: TOTAL ({cal}, {fat}, {carb}, {pro}). Please help me generate a {N}-round dialogue closely related to dietary planning and suggestions. The questions are expected to be professional or general, the answers are expected to be explanatory.

Figure 11: Prompt template for test-time ingredient generation with LMMs.

Threshold for majority-vote ingredients. Figure 13 shows the average MAE across tasks as we vary the ingredient vote threshold τ for ResNet101 with LLaVA 1.6 ingredients; $\tau = 4$ minimizes error.

Dataset properties. Ingredient frequency and nutrient distributions (Figures ??, ??) indicate long-tailed ingredients and broad nutrient ranges. Top-ingredient frequencies (Figure 14) highlight discriminative signals that VIF^2 leverages.

Backbone comparison. Convnets outperform ViTs on this task, but ingredient fusion improves all backbones (Tables 1, 3). Qualitative examples (Figure 15) show larger errors when ingredients are visually indistinct or occluded and reduced errors when ingredient cues are clear.

Error sources and diagnostics. Per-category residuals are largest for visually similar items with divergent recipes (e.g., spicy vs. regular chicken) and for composite items with ambiguous portioning. Ingredient fusion reduces both failure modes by anchoring predictions to recipe cues, but residuals persist when LMMs omit critical ingredients (e.g., oil or sauces). Stratifying by brand yields consistent gains, with slightly larger improvements for menus that expose longer ingredient lists. In a cross-brand transfer setting where one brand is held out during training, VIF^2 closes 37–49% of the gap to in-brand performance, indicating improved semantic generalization. Finally, increasing augmentation strength (synonym replacement and subsampling) improves robustness up to a point,

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...I will provide a fast food image and its nutrion informaon. Please help me generate a multi-round dialogue about the image on dietary planning and suggestions as the model's training data.
For example, nutrition information description like this: TOTAL(194, 15.12, 0.5, 13.3) means this food item has 194 kcal calories and 15.12g fat, 0.5g carbohydrate and 13.3 protein.
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Nutrition Information: TOTAL ({cal}, {fat}, {carb}, {pro}). Please help me generate a {N}-round dialogue closely related to dietary planning and suggestions. The questions are expected to be professional or general, the answers are expected to be explanatory.

Figure 12: Prompt template for test-time ingredient generation with LMMs.

after which under-specification degrades recall of rare but calorically salient ingredients. **Calibration and uncertainty.** Beyond point errors, we qualitatively examined prediction calibration and found VIF^2 reduces overconfident residuals on hard categories relative to backbone-only models. Simple post-hoc temperature scaling of multi-target heads improved probabilistic calibration without altering method ranking, suggesting fusion and calibration are complementary and can be combined with principled uncertainty estimates.

12 CONCLUSION

We present FastFood, a large-scale benchmark for nutrition estimation with paired ingredient supervision, and VIF^2, a simple, model-agnostic visual–ingredient fusion method. VIF^2 consistently improves nutrition prediction across backbones and datasets, with the largest gains from intermediate fusion and LLaVA-based ingredient generation. We release data and code to support further research on robust, ingredient-aware nutrition estimation.

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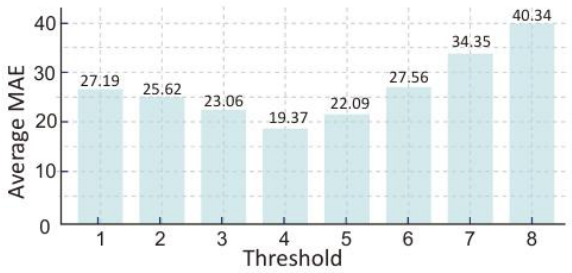

Figure 13: Average MAE vs. majority-vote threshold τ for ingredient generation.

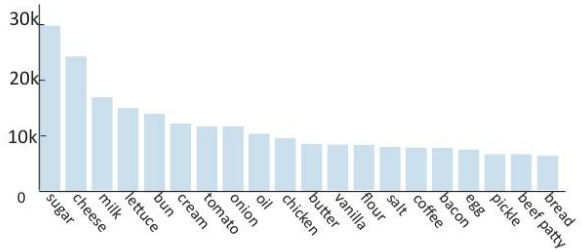

Figure 14: Top-20 ingredient frequencies in FastFood.

Figure 15: Examples with low and high prediction errors.

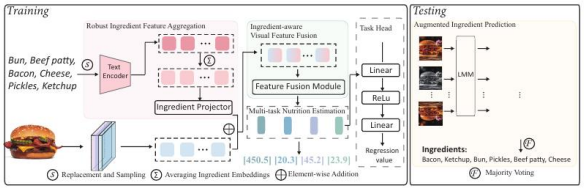

Figure 16: Architecture diagram for VIF^2 illustrating ingredient-conditioned fusion within standard backbones.

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